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Am J Sports Med 2016 44: 1810 originally published online May 10, 2016
DOI: 10.1177/0363546516645082

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Outcomes of Lisfranc Injuries in the National Football League

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Investigation performed at the Hospital of the University of Pennsylvania, Philadelphia, USA

Background: Tarsometatarsal (Lisfranc) joint injuries commonly occur in National Football League (NFL) competition; however, the career effect of these injuries is unknown.

Purpose: To define the time to return to competition for NFL players who sustained Lisfranc injuries and to quantify the effect on athletic performance.

Study Design: Case-control study; Level of evidence, 3.

Methods: Data on NFL players who sustained a Lisfranc injury between 2000 and 2010 were collected for analysis. Outcomes data included time to return to competition, total games played after season of injury, yearly total yards and touchdowns for offensive players, and yearly total tackles, sacks, and interceptions for defensive players. Offensive power ratings (*OPR* = [*total yards*/10] + [*total touchdowns* × 6]) and defensive power ratings (*DPR* = *total tackles* + [*total sacks* × 2] + [*total interceptions* × 2]) were calculated for the injury season and for 3 seasons before and after the injury season. Offensive and defensive control groups consisted of all players without an identified Lisfranc injury who competed in the 2005 season.

Results: The study group was composed of 28 NFL athletes who sustained Lisfranc injuries during the study period, including 11 offensive and 17 defensive players. While 2 of 28 (7.1%) players never returned to the NFL, 26 (92.9%) athletes returned to competition at a median of 11.1 (interquartile range [IQR], 10.3-12.5) months from time of injury and missed a median of 8.5 (IQR, 6.3-13.0) regular-season games. Analysis of pre- and postinjury athletic performance revealed no statistically significant changes after return to sport after Lisfranc injury. The magnitude of change in median OPR and DPR observed in offensive and defensive Lisfranc-injured study groups, −34.8 (IQR, −64.4 to 1.4) and −13.5 (−30.9 to 4.3), respectively, was greater than that observed in offensive and defensive control groups, −18.8 (−52.9 to 31.5) and −5.0 (−22.0 to 14.0), respectively; however, these differences did not reach statistical significance (P = .33 and .21, respectively). Evaluation of the durability of injured players after the season of injury revealed no statistically significant difference in career length compared with controls.

Conclusion: More than 90% of NFL athletes who sustained Lisfranc injuries returned to play in the NFL at a median of 11.1 months from time of injury. Offensive and defensive players experienced a decrease in performance after return from injury that did not reach statistical significance compared with their respective control groups over a similar time period.

Keywords: football; foot; Lisfranc; NFL; return to play

Foot injuries are common in sports, representing more than 15% of all athletic injuries.[8] Although injuries to the tarsometatarsal (Lisfranc) joint account for only 1 in 50,000 orthopaedic traumatic injuries in the general population,[11] approximately one-third of these injuries are caused by low-energy mechanisms, such as athletic activity.[18] American football has been associated with a particularly high frequency of midfoot injuries, with midfoot sprains occurring in 4% of collegiate American football players per year.[12]

Physicians who routinely treat athletic injuries must be equipped to address a unique set of patient concerns. Injured athletes may ask any or all of the following questions: "Will I ever play again? Will I be back in time for playoffs? How will this injury affect my performance after I return?" Answering questions such as these are essential for setting realistic goals and expectations when selecting a plan of care for an athletic injury.

The available literature does not provide a physician with clear data to address the concerns of an athlete regarding return to sport after Lisfranc injury. Tarsometatarsal joint injuries are known to lead to a protracted

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Presented at the 41st annual meeting of the AOSSM; Orlando, Florida; July 2015.

The authors declared that they have no conflicts of interest in the authorship and publication of this contribution.

The American Journal of Sports Medicine, Vol. 44, No. 7
DOI: 10.1177/0363546516645082
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absence from athletic competition; however, the duration of this absence varies significantly in the current literature. In some studies, athletes are reported to return to sport as early as 3 to 4 months after Lisfranc injury.[5,14,17] Other authors report that it is rare for an athlete to return to unrestricted sports activity, particularly those involving twisting and cutting movements, before 6 months after this injury.[13] In addition, no prior study has attempted to quantify the effect of Lisfranc injury on athletic performance in any sport.

Professional American football athletes place unique demands on the injured tarsometatarsal joint that may not be comparable with the study populations examined in the present literature. Currently, there is little information available regarding the time to return to play at this elite level after Lisfranc injury or on the effect of Lisfranc injury on performance after return to the American National Football League (NFL). The purpose of this study was 3-fold: (1) to define the proportion of professional American football players who return to the NFL after sustaining a Lisfranc injury, (2) to define the time to return to competition in the NFL after Lisfranc injury, and (3) to quantify the effect of Lisfranc injuries on athletic performance.

METHODS

Data on NFL players who sustained a Lisfranc injury during a 10-year time period (2000-2010) were collected for analysis. Injuries were identified using NFL game summaries, weekly team injury reports, and player profiles. Players were included in the study group if they sustained a Lisfranc injury between 2000 and 2010 while on the active roster of an NFL team as a running back, wide receiver, tight end, defensive lineman, linebacker, or defensive back. These positions were selected as the focus of the study since components of their performance could be objectively measured. Players were excluded if they had not participated in at least 1 regular season before the season of injury because comparative analysis between pre- and postinjury performance would not be possible. The effect of Lisfranc injuries on the careers of NFL athletes was evaluated by measuring time to return to competition after injury and through player performance analysis.

Return to Play

Return to competition was defined as participation in any NFL regular-season game for at least 1 play. Return to play in practice, preseason games, or in any other professional football league was not considered a return to the same level of competition. Recorded variables included age, experience, position, date of injury, date of return to NFL, games played, and operative versus nonoperative management. Time between date of injury and date of return and number of regular-season games missed due to injury were recorded for all players who returned to the NFL.

Player Performance Analysis

Performance analysis involved comparing the Lisfranc-injured group's athletic performance in 3 seasons before injury with their performance in 3 seasons after injury as well as with a control group over a similar time period. An offensive control group consisted of all running backs, wide receivers, and tight ends without an identified Lisfranc injury who competed in the 2005 season and at least 1 season before and after 2005. A defensive control group similarly consisted of all defensive linemen, linebackers, and defensive backs without an identified Lisfranc injury who competed in the 2005 season and at least 1 season before and after 2005. Performance outcomes data included yearly total yards and touchdowns for offensive players and yearly total tackles, sacks, and interceptions for defensive players. These data were collected for the index season and 3 seasons before and after the index season for all players in the Lisfranc-injured group and the offensive and defensive control groups. The index season was defined as the injury season for those in the Lisfranc-injured group and the 2005 season for those in the control groups.

Athletic performance was evaluated using offensive and defensive power ratings. Offensive power ratings (OPR = [*total yards*/10] + [*total touchdowns* × 6]) and defensive power ratings (DPR = *total tackles* + [*total sacks* × 2] + [*total interceptions* × 2]) were calculated for the index season and for 3 seasons before and after the index season for all players. If a player did not compete in a season during the study period, the power rating for that season was recorded as null (no observation) rather than zero. Players were excluded from performance analysis if their total power rating over 7 seasons was less than 200 points for offensive players or 50 points for defensive players to minimize dilution from players with low production during the study period.

Postinjury durability of Lisfranc-injured athletes was assessed through analysis of total games played after the season of injury for those who returned to the NFL. The number of regular-season games played in all seasons after the season of injury until the 2015-2016 season was recorded for all players in both the offensive and defensive study groups. Similarly, the number of regular-season games played in all seasons after the index season until the 2015-2016 season was recorded for all players in the offensive and defensive control groups described above. These data were then used to determine if Lisfranc-injured athletes experienced a shortened career length compared with controls, as measured by total number of games played.

Power Rating as Outcomes Instrument

The offensive power rating used in this study has been previously validated for use as an outcomes measure for player performance.[2] To include defensive players in our analysis, we evaluated the defensive power rating for reliability and validity. Test-retest reliability was evaluated with intraclass correlation[6]—comparing the variability of

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TABLE 1
Study and Control Group Demographics[a]

	Players With Lisfranc Injury	Controls	*P* Value
Offensive players, n			
No. of players	11	162	
Age, y	27.0 (26.0-28.0)	26.0 (25.0-29.0)	.43
NFL experience (seasons)	5.0 (4.0-6.5)	5.0 (3.0-7.0)	.92
Total exposures (games) in 3 seasons before index year	42.0 (37.5-45.5)	40.0 (29.3-46.0)	.80
Position, RB/WR/TE, n	7/2/2	52/88/22	.04
Defensive players			
No. of players	17	535	
Age, y	27.0 (25.0-30.0)	26.0 (25.0-28.0)	.41
NFL experience (seasons)	6.0 (3.0-9.0)	5.0 (3.0-7.0)	.30
Total exposures (games) in 3 seasons before index year	40.0 (28.0-47.0)	37.0 (24.5-46.0)	.82
Position, DL/LB/DB, n	4/6/7	157/176/202	.87

[a]Values are reported as median (interquartile range) unless otherwise indicated. DB, defensive back; DL, defensive lineman; LB, linebacker; NFL, National Football League; RB, running back; TE, tight end; WR, wide receiver.

different power ratings of the same player with the total variation across all power ratings and all players. The intraclass correlation coefficient, based on correlating power ratings of 535 players for 7 seasons separated by equal time intervals of 1 year, was estimated to be 0.889. For most purposes, values greater than 0.75 may be taken to represent excellent agreement beyond chance.[7] Concurrent criterion-oriented validity[3] was assessed by using a 2-sided *t* test comparing power ratings by Pro Bowl selection for each season of the defensive players studied. For 3016 seasons not associated with Pro Bowl selection, the mean power rating equaled 48.1 ± 32.6. For 179 seasons associated with Pro Bowl selection, the mean power rating equaled 85.4 ± 30.8. This difference was highly statistically significant ($P < .0001$).

Statistical Analysis

Statistical analysis was performed using PASW Statistics version 18.0 (SPSS, Inc). Comparative analyses of independent nonnormally distributed samples were performed using the Mann-Whitney *U* test. $P < .05$ was considered significant. For analyses of player performance, the total *P* value for significance was adjusted using the Bonferroni correction for multiple testing, yielding a threshold value of .001.

RESULTS

Demographics

Lisfranc injuries were identified in 33 NFL athletes in the study period, including 16 offensive players and 17 defensive players. The 1 quarterback and 4 offensive linemen identified to have sustained Lisfranc injuries during our study period were excluded from the study cohort, as these positions do not produce objective performance statistics that are comparable to the remainder of the offensive study group. Of the 11 offensive players who sustained Lisfranc injuries included in our study group, 7 players were running backs, 2 were wide receivers, and 2 were tight ends. The Lisfranc-injured defensive players comprised 4 defensive linemen, 6 linebackers, and 7 defensive backs. Demographic data regarding age, experience in NFL, and total exposures in 3 seasons before injury are listed in Table 1.

Offensive and defensive control groups were created using players who competed in the 2005 season. Of 383 running backs, wide receivers, and tight ends who competed in 2005, 162 players met our inclusion criteria for the control group after we excluded players with previous Lisfranc injuries (n = 11), players who did not compete in at least 1 season before and after 2005 (n = 122), and players who did not earn a total offensive power rating of 200 for the study period (n = 88). Of 854 defensive linemen, linebackers, and defensive backs who competed in 2005, 535 players met our inclusion criteria for the control group after we excluded players with previous Lisfranc injuries (n = 14), players who did not compete in at least 1 season before and after 2005 (n = 276), and players who did not earn a total defensive power rating of 50 for the study period (n = 29). The study and control groups' demographic data, including age, experience in the NFL, and total exposures in 3 seasons before injury, were compared and found to have no statistically significant differences (Table 1).

Return to Play

While 2 of 28 (7.1%) players never returned to the NFL, the remaining 26 (92.9%) athletes returned to competition at a median of 11.1 (interquartile range [IQR], 10.3-12.5) months from time of injury and missed a median of 8.5 (IQR, 6.3-13.0) NFL regular-season games (Table 2). Most players who returned to the NFL (22 of 26; 84.6%) sustained season-ending Lisfranc injuries, returning to the NFL between 9 and 15 months postinjury (Figure 1). There was no statistically significant difference in time to return to play when comparing offensive and defensive players (Table 2). Players treated nonoperatively were

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TABLE 2
Time to Return to the National Football League After Lisfranc Injury by Player and Treatment Type[a]

		Player Type			Treatment Type		
	Overall (N = 28)	Offensive (n = 11)	Defensive (n = 17)	*P* Value	Operative (n = 22)	Nonoperative (n = 6)	*P* Value
Time to return, mo	11.1 (10.3-12.5)	10.9 (10.5-12.4)	11.4 (9.3-12.8)	.81	11.6 (10.7-12.6)	6.2 (1.9-10.7)	.02
Regular-season games missed	8.5 (6.3-13.0)	8.0 (5.5-12.5)	9.5 (6.8-14.3)	.44	10.0 (7.0-13.3)	7.0 (4.5-8.0)	.09

[a]Values are reported as median (interquartile range).

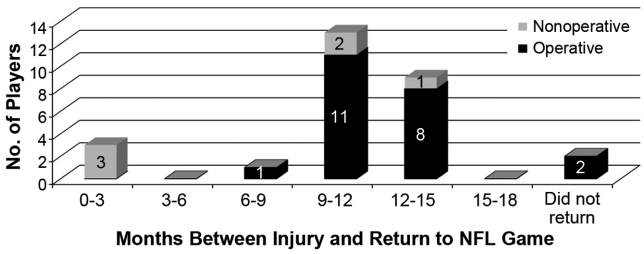

Figure 1. Histogram of grouped frequency distribution of time to return to a National Football League game after a Lisfranc injury by treatment type.

noted to have a trend toward earlier return to play with a median absence from play of 6.2 (IQR, 1.9-10.7) months and 7.0 (IQR, 4.5-8.0) games compared with those treated operatively, who returned after a median of 11.6 (IQR, 10.7-12.6) months (P = .02) and 10.0 (IQR, 7.0-13.3) games missed (P = .09) (Table 2).

Player Performance Analysis

Analysis of pre- and postinjury athletic performance in offensive players revealed no statistically significant change after return to sport after Lisfranc injury. Lisfranc-injured offensive players produced a median OPR per season of 147.9 (IQR, 55.4-178.4) in 3 seasons before injury compared with 59.7 (IQR, 29.2-127.4) in 3 seasons after injury. This produced a median decline in OPR per season of −34.8 (IQR, −64.4 to 1.4). Players in the offensive control group also experienced a decline in power rating over the study period. The offensive controls produced a median OPR per season of 78.9 (IQR, 45.8-134.5) in 3 seasons before the index season and a median OPR of 68.1 (IQR, 41.0-118.5) in 3 seasons after the index season, producing a median decline in OPR per season of −18.8 (IQR, −52.9 to 31.5). While the magnitude of change observed in the Lisfranc-injured group (−34.8) was greater than that in the offensive control group (−18.8), these differences did not reach statistical significance (P = .33) (Figures 2 and 3). This was also true both when comparing each postinjury season power rating with the player's preinjury power rating and when comparing with the corresponding change in performance for the control group (Table 3). Similar results were found on analysis of OPR per game data (Table 3).

Defensive player performance analysis similarly showed no statistically significant change in player production after return to NFL after Lisfranc injury. Defensive players who sustained a Lisfranc injury produced a median DPR per season of 61.0 (IQR, 43.3-85.8) in 3 seasons before injury compared with 48.0 (IQR, 31.8-75.0) in 3 seasons after injury. This produced a median decline in DPR per season of −13.5 (IQR, −30.9 to 4.3). Players in the defensive control group experienced a similar decline in power rating over the study period. The defensive controls produced a median DPR per season of 44.0 (IQR, 23.0-68.0) in 3 seasons before the index season and a median DPR of 40.0 (IQR, 19.8-64.0) in 3 seasons after the index season, producing a median decline in DPR per season of −5.0 (IQR, −22.0 to 14.0). While the magnitude of change observed in the Lisfranc-injured group (–13.5) was again greater than that in the defensive control group (–5.0), these differences did not reach statistical significance (P = .21) (Figures 4 and 5). This was again true both when comparing each postinjury season power rating with the player's preinjury power rating and when comparing with the corresponding change in performance for the control group (Table 4). Similar results were found on analysis of DPR per game data (Table 4).

Analysis of durability, as measured by total number of games played, revealed no statistically significant difference in career length of Lisfranc-injured athletes compared with controls. The offensive Lisfranc-injured study group played in a median of 48 (IQR, 25.25-70.5) regular-season games after the injury season compared with a median of 53 (IQR, 30-82) games after the index season for the offensive control group (P = .61). The defensive Lisfranc-injured study group played in a median of 50 (IQR, 23.75-73) regular-season games after the injury season compared with a median of 46 (IQR, 24-77) games after the index season for the defensive control groups (P = .78).

DISCUSSION

Return to Play

Lisfranc injuries are known to cause a significant absence from athletic competition; however, the duration of this absence varies significantly in the available literature. Curtis et al[5] reported good to excellent results (defined as no significant loss of function from symptoms) in 14 of 19 athletes treated for tarsometatarsal joint complex injuries. While 3 of 19 patients in their study were unable to return to sport,

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TABLE 3
Offensive Player Performance Statistics[a]

	Players With Lisfranc Injury (n = 11)	Controls (n = 162)	*P* Value
OPR per season			
Three pre–index seasons	147.9 (55.4 to 178.4)	78.9 (45.8 to 134.5)	.21
Three post–index seasons	59.7 (29.2 to 127.4)	68.1 (41.0 to 118.5)	.68
Delta	–34.8 (–64.4 to 1.4)	–18.8 (–52.9 to 31.5)	.33
Season 1 pre–index season	142.2 (48.5 to 190.6)	80.2 (48.0 to 154.2)	.28
Season 1 post–index season	125.3 (40.6 to 142.2)	75.1 (35.5 to 129.2)	.51
Delta vs pre–index season	–13.3 (–41.5 to 5.4)	–7.1 (–51.6 to 26.9)	.61
Season 2 post–index	54.8 (19.6 to 187.6)	84.7 (36.6 to 126.6)	.86
Delta vs pre–index season	3.2 (–95.9 to 29.3)	–10.0 (–43.8 to 52.9)	.64
Season 3 post–index season	60.3 (40.0 to 143.8)	65.7 (31.7 to 120.8)	.79
Delta vs pre–index season	–25.4 (–92.1 to 5.6)	–17.2 (–74.0 to 26.6)	.31
OPR per games played			
Three pre–index seasons	10.2 (4.5 to 12.0)	6.2 (3.3 to 8.9)	.13
Three post–index seasons	5.9 (2.3 to 10.1)	5.0 (3.2 to 8.3)	.82
Delta	–1.9 (−4.1 to −0.4)	–1.2 (–3.4 to 1.9)	.33
Season 1 pre–index season	9.7 (3.4 to 12.2)	6.5 (3.8 to 10.0)	.35
Season 1 post–index season	8.8 (3.9 to 11.1)	5.2 (2.7 to 9.2)	.28
Delta vs pre–index season	–0.9 (–1.8 to 1.0)	–0.8 (–3.4 to 1.5)	.99
Season 2 post–index season	6.5 (1.2 to 11.7)	6.0 (3.6 to 8.7)	.85
Delta vs pre–index season	–0.6 (–7.2 to 1.8)	–0.4 (–3.3 to 2.6)	.47
Season 3 post–index season	4.2 (3.5 to 9.0)	5.2 (3.0 to 8.2)	.82
Delta vs pre–index season	–2.2 (–5.5 to 0.9)	–1.4 (–5.1 to 1.8)	.28
Total games played in 3 seasons after injury	35.5 (24.5 to 44.8)	40.0 (28.3 to 45.0)	.72

[a]Values are reported as median (interquartile range) unless otherwise indicated. OPR, offensive power rating. *P* values calculated via the Mann-Whitney *U* test. Threshold for significance *P* = .001 after correction for multiple testing.

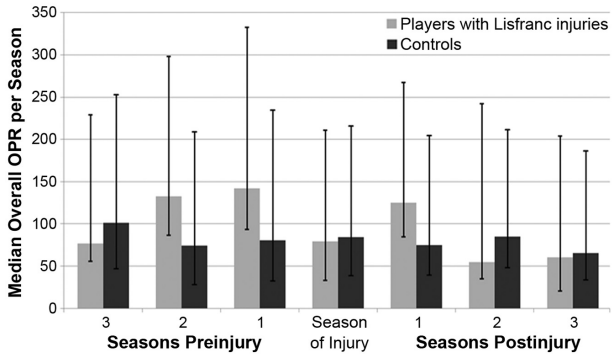

Figure 2. Median offensive power rating (OPR) per National Football League season for Lisfranc-injured study group (gray bars) and offensive control group (black bars). Error bars represent interquartile range.

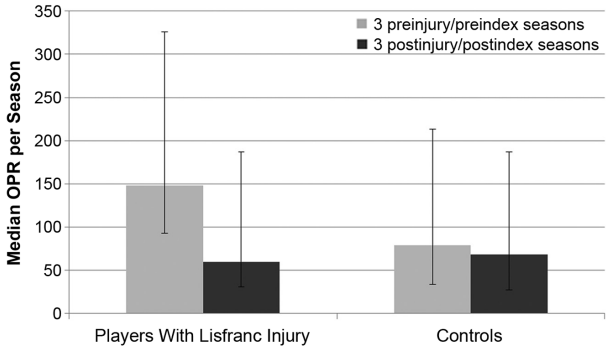

Figure 3. Median offensive power rating (OPR) per National Football League season for 3 pooled pre–index seasons (gray bars) compared with 3 pooled post–index seasons (black bars) for Lisfranc-injured study group and control group. Error bars represent interquartile range.

the remaining 16 patients returned to sport at a mean 4.1 months. None of the patients included in this study sustained their injury playing football, and 14 of the 19 patients in this study were treated nonoperatively. Shapiro et al[17] reported good results of 9 athletes treated for Lisfranc injuries, with all patients returning to competition at a mean 14.7 weeks. Of note, 3 of the 9 patients in their study sustained injuries playing football, and 8 of the 9 patients were managed nonoperatively. Nunley and Vertullo[14] described excellent results (defined as full activity with no pain) in 14 of 15 athletes treated for midfoot sprains. All 15 athletes, including 10 football players, returned to sport at a mean 15.2 weeks. In this study population, 7 of 15 patients were treated nonoperatively.

At this time, only 2 studies have been performed on athletes who have sustained midfoot injuries playing football only, both of which evaluated midfoot sprains in this population. Meyer et al[12] described the outcomes of 23 collegiate football players treated nonoperatively for 24 midfoot sprains, reporting a mean return to practice of

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TABLE 4
Defensive Player Performance Statistics[a]

	Players With Lisfranc Injury (n = 17)	Controls (n = 535)	*P* Value
DPR per season			
Three pre–index seasons	61.0 (43.3 to 85.8)	44.0 (23.0 to 68.0)	.02
Three post–index seasons	48.0 (31.8 to 75.0)	40.0 (19.8 to 68.0)	.36
Delta	–13.5 (–30.9 to 4.3)	–5.0 (–22.0 to 14.0)	.21
Season 1 pre–index season	61.0 (39.5 to 89.0)	46.0 (27.0 to 70.5)	.07
Season 1 post–index season	49.0 (25.3 to 83.3)	42.0 (19.0 to 67.0)	.55
Delta vs pre–index season	–19.5 (–28.3 to 11.3)	–2.0 (–22.0 to 17.0)	.29
Season 2 post–index season	51.0 (33.0 to 75.0)	42.0 (19.0 to 67.0)	.20
Delta vs pre–index season	–9.0 (–21.0 to 11.0)	–4.0 (–27.0 to 16.0)	.75
Season 3 post–index season	49.0 (38.0 to 62.8)	43.0 (23.0 to 68.0)	.48
Delta vs pre–index season	2.0 (–38.0 to 12.0)	–5.0 (–26.0 to 15.0)	.81
DPR per game played			
Three pre–index seasons	4.5 (3.4 to 5.7)	3.1 (1.8 to 4.7)	.01
Three post–index seasons	3.8 (2.4 to 4.8)	3.0 (1.7 to 4.4)	.23
Delta	–1.1 (–2.0 to 0.6)	–0.2 (–1.2 to 0.8)	.15
Season 1 pre–index season	3.9 (2.5 to 6.4)	3.3 (1.8 to 5.0)	.11
Season 1 post–index season	4.5 (2.4 to 5.3)	3.1 (1.6 to 4.8)	.19
Delta vs pre–index season	–0.6 (–1.5 to 1.2)	–0.2 (–1.3 to 1.1)	.60
Season 2 post–index season	4.4 (2.9 to 5.3)	3.2 (1.7 to 4.6)	.12
Delta vs pre–index season	–0.1 (–1.0 to 0.6)	–0.3 (–1.5 to 0.9)	.98
Season 3 post–index season	3.1 (2.8 to 4.1)	3.3 (1.9 to 4.7)	.86
Delta vs pre–index season	0.1 (–2.1 to 0.5)	–0.3 (–1.4 to 0.8)	.67
Total games played in 3 seasons after injury	37.5 (23.8 to 45.5)	38.0 (24.0 to 35.0)	.77

[a]Values are reported as median (interquartile range) unless otherwise indicated. DPR, defensive power rating. *P* values calculated via the Mann-Whitney *U* test. Threshold for significance *P* = .001 after correction for multiple testing.

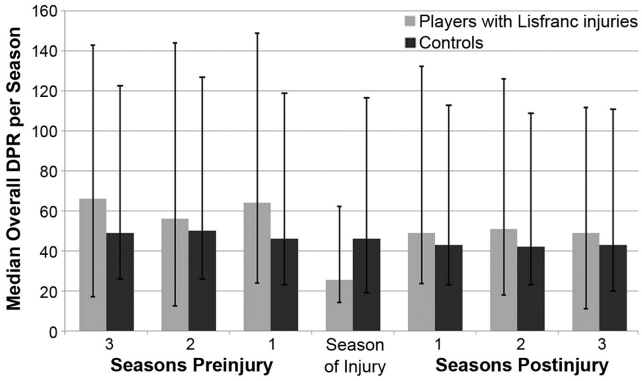

Figure 4. Median defensive power rating (DPR) per National Football League season for Lisfranc-injured study group (gray bars) and defensive control group (black bars). Error bars represent interquartile range.

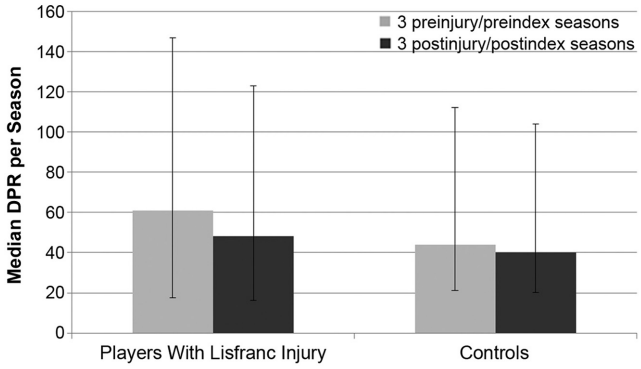

Figure 5. Median defensive power rating (DPR) per National Football League season for 3 pre–index seasons (gray bars) compared with 3 post–index seasons (black bars) for Lisfranc-injured study group and offensive control group. Error bars represent interquartile range.

13.8 days and a mean time to "full healing" of 40.5 days. Osbahr et al[15] reported the outcomes of 15 NFL players treated for midfoot sprains. Nonoperative management of 12 players with grade 1 and 2 midfoot sprains resulted in a mean 11.7 days of time lost from participation, while operative management of the remaining 3 players with grade 3 sprains resulted in a return after 73 days for 1 player and season-ending injuries in 2 players.

In our study, 26 of 28 (92.9%) NFL athletes who sustained Lisfranc injuries returned to competition in the NFL. With the exception of the 2 (7.1%) players who never returned to the NFL, the remaining 26 (92.9%) athletes returned to play at a median of 11.1 (IQR, 10.3-12.5) months from time of injury and missed a median of 8.5 (IQR, 6.3-13.0) NFL regular-season games. Most players who returned to the NFL (n = 22 of 26; 84.6%) sustained season-ending Lisfranc injuries, returning to the competition between 9 and 15 months after injury. Only 3 players (11.5%) returned to play during their injury season, with 1

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additional player returning midway through the season after an offseason injury (Figure 1).

Our reported median 11.1-month absence from sport is significantly longer than those reported in the previously mentioned studies on Lisfranc injuries in athletes.[3,12,14,15,17] However, other authors have reported longer periods of restricted athletic activities after tarsometatarsal joint injuries that are consistent with the findings of our study. Myerson and Cerrato[13] stated that it is rare for an athlete to return to unrestricted sports activity, particularly those involving twisting and cutting movements, before 6 months after injury. Our study was limited to players who were reported to have sustained a "Lisfranc injury" on NFL injury reports and excluded those listed with a "midfoot sprain." This likely resulted in the exclusion of many athletes with less severe midfoot injuries who would have met the criteria of the studies by Meyer et al[12] and Osbahr et al.[15] In addition, our definition of return to play, the participation in any American NFL regular-season game for at least 1 play, complicates making direct comparisons with the results of these prior studies that did not require return to regular-season game play as their endpoint. Finally, NFL athletes place unique demands on the injured tarsometatarsal joint that may not be comparable with the study populations examined in most of the present literature.

NFL players in this study treated nonoperatively were noted to have a trend toward earlier return to play with a median absence from play of 6.2 (IQR, 1.9-10.7) months and 7.0 (IQR, 4.5-8.0) games compared with those treated operatively, who returned after a median of 11.6 (IQR, 10.7-12.6) months ($P = .02$) and 10.0 (7.0-13.3) games missed ($P = .09$). While it is possible that surgical intervention may result in a longer time to return, this is more likely a potential surrogate for greater injury severity. The high proportion of injuries requiring surgical intervention in our study group (n = 22) compared with those managed nonoperatively (n = 6) also may have contributed to the longer time to return to play found in this study compared with the results previously reported in the literature, which featured higher rates of nonsurgical management.[3,10,12,15,17]

To our knowledge, no previous study has quantified the effect of Lisfranc injuries on athletic performance after return to competition. Curtis et al[5] stated that 2 of 16 patients who returned to sport in their study had to change or limit their athletic activities due to midfoot pain, but the authors offered no further analysis of athletic performance after Lisfranc injury for the remainder of their study population. Shapiro et al[17] reported that all 9 athletes in their study returned to "preinjury levels" of competition but did not comment on the quality of performance at this level after injury. The studies by Nunley and Vertullo[14] and Meyer et al[12] both provided no description of postinjury athletic performance. Osbahr et al[15] concluded that "it was not possible to differentiate the specific return-to-play level related to preinjury performance level" in their study population.

The player performance analysis performed in this study aimed to quantify the effect of Lisfranc injuries on performance of NFL athletes. On initial observation of the data collected in this study, an improvement in performance of the offensive study group was noted that peaked 1 season before the index season and was higher than that of the corresponding offensive control group (Figure 2). This higher production pre–index season would be consistent with more on-field time with greater exposure to potential injury-causing plays. After the injury season, the offensive study group experienced a greater decline in performance than that experienced by the corresponding offensive control group (Figure 2). However, on statistical analysis, this postinjury decline in performance did not reach statistical significance both when comparing each postinjury season OPR with the OPR of 1 season preinjury (most representative of preinjury level of play) and when comparing with the corresponding change in performance for the offensive control group over a similar time period (Table 3). As a result, the magnitude of change in median OPR observed in the Lisfranc-injured offensive study group, –34.8 (IQR, –64.4 to 1.4), was greater than that observed in the offensive control group, –18.8 (IQR, –52.9 to 31.5) (Figure 3); however, these differences did not reach statistical significance ($P = .33$) (Table 3).

Player performance analysis for the defensive players revealed similar findings to the offensive player performance analysis. The defensive study group also performed at a higher level than its corresponding control group before the index season and experienced a greater decline in performance after the injury season than that experienced by the controls (Figures 4 and 5). However, on statistical analysis, this postinjury decline in performance did not reach statistical significance both when comparing each postinjury season DPR with the DPR of 1 season preinjury and when comparing with the corresponding change in performance for the control group over a similar time period (Table 4). As a result, the magnitude of change observed in the Lisfranc-injured defensive study group, –13.5 (IQR, –30.9 to 4.3), was greater than that observed in the defensive control group, –5.0 (IQR, –22.0 to 14.0); however, these differences also did not reach statistical significance ($P = .21$).

Analysis of athletic performance in both offensive and defensive players revealed declines in performance after return to sport after Lisfranc injury that did not reach statistical significance. Based on these data, it is possible that Lisfranc injuries may result in a mild decline in player production after return to the NFL that this study was underpowered to detect due to small sample sizes. Of note, 14 of 28 (50%) players in the Lisfranc injured-study group produced an OPR or DPR in at least 1 postinjury season that exceeded their preinjury level of play (OPR/DPR of 1 season preinjury), demonstrating that a return to preinjury performance is possible for NFL players after Lisfranc injuries. Further study is needed to identify specific factors associated with poor prognoses.

Analysis of durability after Lisfranc injury, as measured by total number of games played, revealed no statistically significant difference in career length of Lisfranc-injured athletes compared with controls. The offensive and defensive Lisfranc-injured study groups played in a median of 48 (IQR, 25.25-70.5) and 50 (IQR, 23.75-73) regular-season games after the injury season, respectively, compared with a median of 53 (IQR, 30-82) and 46 (IQR, 24-77) games after the index season for the offensive and defensive

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control groups, respectively (P = .61 and .78, respectively). Based on these data, it does not appear that Lisfranc injuries alter the duration of NFL careers for those who return to play at this level; however, it is possible that a difference may exist that this study was underpowered to detect due to small sample sizes.

This study has limitations. First, this study may be affected by a selection bias due to our reliance on NFL game summaries, weekly team injury reports, and player profiles to select the study population without access to physical examination findings or imaging studies to confirm the reported diagnosis of Lisfranc injury. Similarly, this study did not evaluate the effect that severity of injury may have on both time to return to play and performance of NFL athletes after Lisfranc injury. This study also did not assess the role that specific treatment regimens, such as different nonoperative rehabilitation programs or various modes of surgical fixation, may play in expediting a player's return to the NFL and optimizing postinjury performance. These limitations may be of particular significance for the study of Lisfranc injury outcomes due to the intrinsic spectrum of severity and more variable treatment options for Lisfranc injuries compared with other athletic injuries, such as anterior cruciate ligament ruptures. A variety of surgical techniques for the treatment of Lisfranc injuries have been described in the literature, including closed reduction with percutaneous screw fixation, open reduction and internal fixation with screw fixation or dorsal plating, primary arthrodesis, and utilization of suture-button fixation constructs.[1,4,9,10,16] There are limited data available regarding the outcomes of specific surgical techniques with regard to athletic performance. Further research on these variables is warranted to provide additional prognostic information regarding Lisfranc injuries in NFL athletes.

This study used power ratings based on professional athlete game performance as measures of treatment outcomes. Inherent limitations exist when representing human performance with a single value. Power ratings are not fully comprehensive, as many attributes essential to player performance, such as strength, speed, experience, and leadership, were not included in the calculation of these ratings. However, the power ratings were defined by the weighted sums of total yards and touchdowns for offensive players and total tackles, sacks, and interceptions for defensive players, arguably the most important statistics for these positions. Both the offensive and defensive power ratings were proven to be statistically reliable and valid as outcomes measures and were responsive to both Lisfranc injury and treatment. In addition, the data to calculate power ratings were readily available and consistently recorded, facilitating the tracking of player performance each year without losing subjects to follow-up.

CONCLUSION

More than 90% of NFL athletes who sustained Lisfranc injuries returned to competition in the NFL after injury. Of those who returned, approximately 85% missed the remainder of their injury season, returning to play at a median 11.1 months from time of injury. Offensive and defensive players experienced a decrease in performance after return from injury that did not reach statistical significance compared with their preinjury level of performance and compared with control groups over the same time period. These data suggest a return to preinjury performance is possible for NFL players after Lisfranc injuries. This information may assist in providing realistic patient expectations after Lisfranc injuries, particularly when discussing implications of future athletic performance with athletes and coaches.

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